

Visionary

→ 2026 BEHAVIOR · per-primitive diversity floor · NO Discord/Drive

Forward claim

Treat Lin'25 32×50 as the single-skill diversity unit for BEHAVIOR-scale household robotics.
Spend a ~20k global demo pool on covering units + shared generalist prior — don't deepen easy pairs.

20k mapping = TENTATIVE · recipe = CLEAR

BEHAVIOR connection

Many household tasks × house-scale scenes × object variation

Need

Per-primitive coverage
before more demos
of the same pair

Floor

≥ 32 (scene, object)
pairs × ~50 demos
before deepening

Meter

Lin'25 measures
how much churn
is enough

Complement

DROID scene-churn
+ UMI handheld
as collection engines

Per-primitive diversity floor — 32×50

CLEAR recipe · use as lower bound per skill before depth

32

pairs / primitive

$\times 50$

demos / pair

$\approx 1.6k$

demos / skill floor

$\sim 90\%$

ZS target (paper tasks)

Operational rule

Before buying more demos of the same (scene, object), hit ~ 32 pairs $\times \sim 50$. Multi-obj/env only helps when $\#envs$ is small (Fig.6) — at $M \geq 16$, 1 obj/env $\approx$ multi-obj.

Dexterous / multi-minute tasks may need $K > 50$ (authors) — treat 50 as similar-difficulty default, not universal constant.

~20k global budget — TENTATIVE mapping

Aligns OXE visionary pack framing · not a paper number

Naive equal split fails

If ~100 tasks share ~20k demos → ≈ 200 demos/task — far below this paper's ~1600 if each task is its own policy.

(a) Share priors

$\pi_{0.5}$ / GR00T / OpenVLA-class
so per-task demos can be fewer
than 32×50

(b) Concentrate

Spend full 32×50 on hardest
contact-rich tasks; transfer
for easier primitives

(c) Hybrid

Generalist backbone + diversity
cards on hard skills; forbid
 $K > 50$ on easy pairs

Follow-up research agenda

What to measure next — falsifiable

1 Heads Repeat Fig.5 with ACT / π_0 -FM / VLA — do exponents move?

2 Task axis Fit laws on task count + language (OpenVLA/Octo axis).

3 RL atop IL RL fine-tune scaling on top of IL diversity laws (§7 ask).

4 Embodiment Does M_{robot} enter the same power family as M_{env} ?

5 1191 verify Measure Mouse 0.99 extrapolation or estimate Y_∞ with denser grid.

6 Hard skills Dexterous / deformable / long-horizon K thresholds.

Collector GUI — force new pairs before $K > 50$

Productize the diversity $\gg$ depth moral

1 Quota

GUI refuses additional demos on a pair once $K \geq 50$ until a new (env,obj) is registered.

2 Diversity card

Every session logs env_id / obj_id / pair_id — mix card visible to leads.

3 Hard-skill flag

Override $K > 50$ only with explicit “dexterous/LH” tag + reason.

4 SKU

Data marketplace sells “32-pair skill packs,” not “10k demo dumps.”

New apps

Beyond the paper's four tasks

Eval harness

Blind multi-checkpoint scoring protocol as standard for scaling claims.

Sim BEHAVIOR

Privileged state expands pair diversity cheaply; real K=50 polish.

Per-env QA gate

Ship only if min-env SR $\geq$ threshold — kill mean-only dashboards.

Capacity budget

Spend GPU on ViT scale + FT, not U-Net width — Tab.2 moral.

Mix manifest

Versioned diversity cards next to every HF shard (DROID-aligned).

Skill pack SKU

Commercial 32×50 packs with held-out env test certificate.

Anti-patterns — Visionary forbids

NO Discord / Drive content in this deck · stay on paper → BEHAVIOR

Deepen easy pairs Dumping $K \gg 50$ on solved (env,obj) while hard primitives starve.

Mean-only SR Shipping on average ~90% while one env is 40%.

MSE as ship gate LoRA trap — real score can invert MSE ranking.

Silent head swap Assuming ACT/ π /VLA inherit DP exponents without re-fitting Fig.5.

OXE replacement myth Treating Lin'25 as foundation-mix solution — it isn't.

Invented Discord lore No Discord/Drive citations — paper + CLEARpacks only.

Motion literacy for BEHAVIOR entries

DP chunks + ACT TE @ 5 Hz · horizon 16 — carry into challenge policies

DP / FM / VLA

choose head; re-fit laws

ACT TE

stitch chunks

Hor 16

@ ~5–15 Hz class

Obs 2-3

history for ambiguity

Diversity card

M/N/K logged always

Blind eval

multi-ckpt protocol

Narrative map

OXE = who · DROID = where · Lin'25 = how much per skill · BEHAVIOR = progress

OXE

WHO
embodiments
labs

DROID

WHERE
scenes
protocol

Lin'25

HOW MUCH
32×50 / skill
power laws

UMI

HOW COLLECT
handheld
in-wild

$\pi_{0.5}$ / GROOT

SHARED PRIOR
fewer demos
per task OK

BEHAVIOR'26

PROGRESS
household
coverage

Numbers tattoo — pin these

Every deck carries the same spine numbers

diversity $\gg$ depth

envs/objs/pairs $\gg$ K

32 $\times$ 50

pairs $\times$ demos $\rightarrow$ $\sim$ 90%

85-92.5%

Tab.1 SR $\cdot$ 4 tasks

$\alpha \approx -0.47...-0.84$

Fig.5 exponents

0.66 / 0.81 / 0.90

ViT-S / B / L

0.88 / 0.90 / 0.83

U-Net S / base / L

0.00 / 0.03

frozen / LfS fail

K $\approx$ 50

plateau / pair

>40k / >15k

demos / rollouts

1191 pairs

MA score 0.99 extrap.

DP + ACT TE

@5 Hz $\cdot$ horizon 16

FT 0.90 / LoRA 0.72

full FT required

Exam bite — Visionary

EXAM BITE

For BEHAVIOR, treat Lin'25 32×50 as the single-skill diversity unit; spend the ~20k pool on covering units + shared generalist prior — don't deepen easy pairs. Share priors ($\pi_{0.5}$) or concentrate on hard tasks. Collector GUI: force new pairs before $K > 50$.

Visionary — close

Takeaways + exam bite

1. BEHAVIOR floor: 32×50 per primitive before deepening (CLEAR recipe).
2. $\sim 20k$ global mapping TENTATIVE — share priors or concentrate on hard tasks.
3. Follow-ups: re-fit Fig.5 on ACT/ π /VLA; task-axis; verify 1191; GUI force new pairs.
4. NO Discord/Drive — paper + CLEAR packs only. Tattoo: diversity $\gg$ depth.

EXAM BITE

BEHAVIOR: Lin'25 32×50 = single-skill diversity unit; cover units + shared prior — don't deepen easy pairs.